

EMRE BATU ATAKAN

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EDUCATION

PRIVATE BEŞTEPE ANATOLIAN HIGH SCHOOL

Ankara, Türkiye | 2020 - 2025

Full scholarship - Merit-based educational scholarship for placing in the top 1% in the national high school entrance exam.

- National high school entrance exam – top 0.47% in ~1.5 million
- 11th grade in high school – grade average 98.34 / 100.00
- IGCSE Grades: A* in Chemistry, Mathematics, Physics, English as a Second Language, Turkish; A in Biology
- A Level Grades: A in Mathematics, Physics, and Chemistry
- The Digital SAT – 1530/1600 superscore

INTERNSHIPS

[THRESH POWER](#) - SOFTWARE DEVELOPMENT INTERN

Summer 2024

- Thresh Power is a San Francisco startup that streamlines M&A processes using machine learning.
- Worked on OpenAI API integration for document summarization and Q&A. Developed benchmarks to compare Large Language Model (LLM) model performance. Build full-stack product features.
- Built and deployed a Python microservice on GCP. The service uses OpenAI assistants API to generate due diligence reports on user-provided files. Implemented separate staging and production environments for safer testing.
- Evaluated Ragas and various metrics as a means of understanding the performance impact of different prompting strategies. Documented findings and criticisms of different methodologies.
- Shipped streaming file downloading feature as a React component with a corresponding Python API. Implemented 100% unit test coverage for the Python API. The feature is used by the customers every day.

COMPETITIONS & AWARDS

FIRST ROBOTICS COMPETITION (FRC)

2022 - Present

- FIRST Robotics Competition ([FRC](#)) is an international high school robotics competition.
- 2022 Bosphorus Regional – Rookie Inspiration
- 2022 Pendik Off-Season – Creativity Award
- 2023 Bosphorus Regional – Team Sustainability Award

AMERICAN MATHEMATICS OLYMPIADS: AMO 2024

2024

- National 7th Place / Global 269th Place

CARIBOU CONTESTS

2022 - 2024

- The Caribou Mathematics Competition is an international online math contest held six times over the school year covering topics such as algebra, logic, and geometry.
- International Caribou Contest 2022-2023 – 1st out of 8,709
- International Caribou Cup 2022-2023 – 31st out of 8,709
- International Caribou Contest 2023-2024 – 7th out of 2,397
- International Caribou Cup 2023-2024 – 16th out of 2,397

UNIVERSITY OF WATERLOO MATHEMATICS CONTEST

2023 - 2024

- Cayley Math Contest and Fermat Math Contest are international contests that the University of Waterloo organizes for high school students to attend every year.
- Cayley Math Contest 2023 - Distinction, Top 25%
- Fermat Math Contest 2024 - Honor Roll, Distinction, Top 25%

PURPLE COMET! MATH MEET

2023

- National 2nd place out of a total of 87 national teams.

[TZY](#) (Türkiye Logic Competition) CHAMPIONSHIP

2019 - 2023

- Finals – National 14th place (2019 - 2020)
- Regional Finals Qualified (2021 - 2022)
- Regional Finals Qualified (2022 - 2023)

INTERNATIONAL STEM OLYMPIAD

2022 - 2023

- Mathematics - World 14th place

PRIVATE ARI SCHOOLS 12TH MATHEMATICS OLYMPIAD

2020

- Gold Medal and Honorable Mention

TURKISH BRAIN OLYMPIADS

2020

- Regional Finalist

PRIVATE MAYA SCHOOLS MATHEMATICS OLYMPIAD

2018 - 2019

- Private MAYA Schools Mathematics Olympiad is a national competition that is held annually. Schools in Türkiye create their own teams to compete in this competition.
- 1st place (2019)
- 2nd place (2018)

INTERNATIONAL KANGAROO MATHEMATICS COMPETITION

2018

- The International Kangaroo Mathematics Competition takes place annually and more than 6 million people compete. In Türkiye, more than 100 thousand students attend every year.
- National 9th place

NATIONAL BILSEMFEST

2017

- Robotic Surprise Mission Category - National 3rd place

PROJECTS

TÜBİTAK STUDENT RESEARCH PROJECT COMPETITION

2020 - Present

TÜBİTAK Student Research Project Competitions are national research competitions that focus on different subjects. Participated in the software development track.

Third Eye (2020 - 2021)

- Third Eye is a data protection software based on machine learning and face detection. The goal is to detect potential intruders looking at the screen and/or reading from the screen. If intruders are detected, the software locks the device screen, puts up a passcode verification, and alerts the user. Different user profiles can be created to prevent the software from locking unexpectedly. Python and OpenCV libraries are used for the implementation.
- Worked on both back-end and front-end development processes. Designed a user interface and created it using Python Tkinter added a personal notifier to alert the user and coded a face detection and identification system using machine learning with Python OpenCV.

GÖKRATES (Göz Kırpma Testinden Rahatsızlık Tespit Sistemi) (2022 - 2023)

- GÖKRATES is a healthcare software that is based on machine learning and face detection. The main goal of this software is to detect the trends of the user's blinking and squinting over time while using their devices. After working over the trends, the software suggests possible solutions using its database which is built with the help of experts in the field. Different user profiles can be created to collect personalized statistics. Python and OpenCV library are used for the implementation as well as Google's machine learning platform MediaPipe for model training purposes.
- I worked on both back-end and front-end development processes. Designed a user interface and created it using Python Pyqt along with coding a face detection and identification system using machine learning with Python OpenCV with the implementation of Google's machine learning platform MediaPipe to train models. Moreover, worked on creating a suitable algorithm to minimize the effects of distance from the user's video input devices during eye detection.

WORLD ROBOT OLYMPIAD - WRO

2023

AFES (Autonomous Fire Extinguisher System) (2023)

- AFES's goal is to detect and act early on cargo ship fires before the help arrives. The goal is to minimize expected harm. Various circuit elements, such as ESP32, are used in the process of building the project and C++ is used in coding.

EXTRACURRICULAR

LEADERSHIP

2022 - Present

- FRC Team Captain of The Chaotics 8795 – 2023-2024 season. Worked for both the robotics and the public relations departments of the team. Created sub-groups for departments of the team such as 3D design, electrical, and coding. Led a total of 122 team members successfully and won various awards.
- Math Olympiad Club Leader. Created a curriculum with the help of math tutors to create both national and international teams to compete in competitions. Led 58 people in the club and accomplished success in competitions such as the Caribou Contest and International Stem Olympiad.
- BestMUN Head of Admin. I found funding for the MUN (Model United Nations) conference my high school team organized. Led a team of 12 people throughout the conference for four days. Successfully conducted the conference.

- FLL (FIRST Lego League) Competition Organizer. Prepared my high school for the competition day. Led a total of 15 students who were associated with the FRC team. The competition took 2 days, and awards successfully found their owners.

DUKE OF EDINBURGH INTERNATIONAL AWARD

2022 - Present

The Duke of Edinburgh International Award is an education framework operating in more than 130 countries and territories to help young people transform their lives and communities.

Gold Certificate

- Gave one-hour-long lessons for fifty-two weeks to middle school students who were preparing for the national high school entrance exam
- Prepared a fitness program that mainly revolved around jogging, and followed the plan for seventy-eight weeks
- Joined a three-day workshop and made sculptures out of scraps for people with disabilities

ABRSM PIANO

2019 - 2023

Grade 4 (2019)

- Score: Merit

Grade 6 (2021)

- Score: Merit

Grade 8 (2023)

- Score: Merit

RESEARCH STUDENT

2015 - Present

BILSEM provides advanced education and research opportunities for gifted students. To be accepted into BILSEM, students selected from their schools undergo a two-part examination process. The first part consists of a written examination. Only the students above the minimum passing score move on to the second part, which consists of interviews with experts and tutors. Only the students who perform well during interviews get into BILSEM.

- Took classes on Geometry, Calculus, Algebra, Combinatorics, Applied Mathematics, Number Theory, Computer Science, and Cybersecurity (2015 - Present)
- Worked on TÜBİTAK projects (2020 - Present)

UYEK CONFERENCE

2023

- UYEK Conference aims to point out the issues about the education of gifted students, what has been done, and what further changes can be made to improve the conditions and learning experience.
- Participated as a speaker, and presented the impacts of the integration of AI into the education of gifted students, supporting the ideas presented with previous studies and projects conducted by me in BILSEM.
- These projects included a simple face recognition system that my team and I had developed. The System's final goal was to create a summarized report of the reactions acquired from the students.

SKILLS

Technical Skills: Proficient in Python, SOLIDWORKS, Java, C++, Arduino

- Python - worked on several projects, focusing on machine learning and artificial intelligence projects: face recognition and face landmark detection systems. Attended Tübitak Student Research Project competitions using Python.
- Used SOLIDWORKS to create robot design of FRC Team The Chaotics 8795. Also, worked with Java to write the robot code.
- Built a cargo ship fire prevention robot from the ground up using ESP32 Wi-Fi and Bluetooth MCU, Arduino UNO, MQ2 flammable gases and smoke sensor, MQ7 carbon monoxide sensor, MQ9 carbon monoxide and flammable gases sensor, MQ135 air quality measurement sensor, L298N motor driver, flame sensor, and many other circuit components. Used C++ to program the robot logic.

Soft Skills: Leadership, Teamwork, Problem-solving, Time management

Languages: Fluent in English and Turkish; beginner-level German and Russian